

Subspeciation in the Meadow Mouse, *Microtus montanus*, in Wyoming and Colorado

Sydney Anderson



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Author: Sydney Anderson

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MONTANUS, IN WYOMING AND COLORADO ***

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Microtus montanus reaches the eastern limits of its geographic distribution in Wyoming and Colorado. There the mountains, but in general not the lowlands, are occupied by this species. A certain minimum of moisture may be of direct importance to the mouse and certainly is indirectly important, because certain hydrophytic or mesophytic grasses used by the mouse for food, for protection from enemies, and for shelter from the elements are dependent on the moisture. Areas suitable for *Microtus montanus* are separated by deserts that are dominated by sagebrush and other xerophytic plants or by forests or rocky exposures at higher altitudes. A relatively small percentage, probably less than ten per cent, of the total area even in the more favorable parts of the range of the species is suitable for occupancy. In these mice, as in other microtines (Elton, 1942; Piper, 1909), there are seasonal, and irregularly multiannual fluctuations in population density, which sometimes are extreme. Consequently the mice at some times seem to be absent from suitable habitats, and at some other times occur there in amazingly large numbers.

Because the species is broken up into partly isolated, or at times completely isolated, colonies or local populations it may be supposed that various evolutionary forces such as selection and random genetic drift operate to foster variation. The degree to which racial distinction is attained may depend upon these forces and the time available. In *Microtus montanus* in the eastern Rocky Mountains the degree of subspecific distinction is not great.

The study here reported upon is based on 1,187 specimens of *Microtus montanus* from Wyoming, Colorado, Idaho and Montana, and on work in the field. I spent approximately four months in the field in this area, in the summers of 1950, 1951, and 1952. The specimens studied were arranged according to localities and the larger series were compared statistically. Each of two series, totaling 136 specimens, was studied intensively to ascertain the kind and range of variation within single populations. Twenty-seven measurements, various proportions based on these measurements,

and differences in color were analyzed. Fifteen characters, judged to be most significant, were selected for use in comparing all series. In addition, certain characters that can not be expressed easily by measurements, such as inflation of the auditory bullae and the curvature of the zygomatic arch, were observed. The studies by A. B. Howell (1924) of variation in *Microtus montanus yosemite* Grinnell in California and those by O. B. Goin (1943) of *Microtus pennsylvanicus pennsylvanicus* (Ord) were useful. The external measurements are from the collectors' field labels. The measurements of the skull all were taken with dial calipers reading to a tenth of a millimeter. The anteroposterior measurements of the skull all were taken along the shortest line between the points specified below and are not necessarily along a line parallel to the long axis of the skull. These measurements were taken on the left side of the skull whenever possible. Some of the skulls are damaged and therefore some measurements could not be taken and are omitted in the computations. Measurements are in millimeters.

The results of these studies were submitted to the Department of Zoology and the Graduate School of the University of Kansas in partial fulfillment of the requirements for the degree of Master of Arts (1952) and are available in manuscript form at the Museum of Natural History and the library of the University of Kansas.

EXPLANATION OF MEASUREMENTS

Caudal index.—the length of the tail expressed as a percentage of the length of the head and body. The length of the head and body is the collector's measurement of total length less the length of the tail.

CRANIAL MEASUREMENTS OF LENGTH.

Condylbasilar length.—from the exoccipital condyle to the most posterior point on the border of the alveolus of the upper incisor.

Alveolobasilar length.—from the posterior border of the alveolus of the third upper molar to the posterior border of the alveolus of the incisor.

Palatilar length.—from the anteriormost part of the posterior border of the bony shelf of the palate to the posteriormost part of the alveolus of the incisor.

Alveolar length of upper molar tooth-row.—from the most posterior point of the alveolus of the third upper molar to the most anterior point of the alveolus of the first upper molar.

MEASUREMENTS OF BREADTH.

Zygomatic breadth.—greatest transverse width.

Interorbital breadth.—the breadth of the interorbital constriction.

Lambdoidal breadth.—between the lateralmost points of the lambdoidal crest.

Prelambdoidal breadth.—between the medialmost margins of the prominent fenestrae in the posterodorsal parts of the squamosal bones. To these fenestrae Howell (1924:995) applied the adjective "prelambdoidal," but other authors have used other names (see Hill, 1935:127).

Depth of braincase.—shortest distance from the ventral side of the cranium at the suture between the basioccipital and basisphenoid bones to the dorsal surface of the cranium (usually not perpendicular to the long axis of the skull).

The history of our knowledge of *Microtus montanus* in this area begins with the early work of the United States Bureau of Biological Survey directed by C. H. Merriam (1891), and participated in by Vernon Bailey (1900, 1917), Merritt Cary (1911, 1917), and others. The changes in nomenclature which grew out of increased understanding of these mice through additional collecting are expressed in the synonymies under the accounts of subspecies. As a result of my studies two of the three subspecific names previously proposed for mice from this area have been retained although changes are proposed in the ranges assigned to the two subspecies and two additional heretofore unrecognized subspecies are named and described. Furthermore the additional specimens and my studies of variation make

modifications in the characterization of these subspecies necessary. Not all of the samples here assigned to a single subspecies are identical and I therefore list and discuss some of the local variants.

Numerous members of summer field parties from the Museum of Natural History at the University of Kansas collected most of the specimens studied and wrote field notes that have been helpful. I am grateful to these persons and to Professor E. R. Hall and Dr. R. H. Baker for their assistance and helpful suggestions. Specimens in the following museums were made available by their respective curators: Chicago Natural History Museum by Mr. Colin C. Sandborn, The Museum of Zoology at the University of Michigan by Dr. E. T. Hooper, The American Museum of Natural History by Mr. G. G. Goodwin, The United States National Museum by Dr. David H. Johnson and the Biological Surveys Collection by Miss Viola S. Schantz. A fellowship from the National Science Foundation made possible the studies at the museums other than at the University of Kansas.

***Microtus montanus nanus* (Merriam)**

Arvicola (Mynomes) nanus Merriam, N. Amer. Fauna, 5:62, July 30, 1891.

Microtus montanus nanus, Hall, Proc. Biol. Soc. Washington, 51:131, August 23, 1938.

Microtus nanus, Bailey, N. Amer. Fauna, 17:30, June 6, 1900 (part).

Microtus montanus caryi Bailey, Proc. Biol. Soc. Washington, 30:29, February 21, 1917.

Microtus nanus nanus, Cary, N. Amer. Fauna, 42:43, October 3, 1917 (part).

Type.—Adult male No. 23853/31253, U. S. National Museum, Department of Agriculture collection, from Pahsimeroi Mountains, Custer County, Idaho; obtained by C. Hart Merriam and Vernon Bailey, September 16, 1890.

Range.—Idaho; southwestern Montana; most of the southwestern half of Wyoming; southward to central Colorado. See [figure 1](#).

Comparisons.—Comparisons with subspecies named as new in this paper will be found in the accounts of those subspecies beyond. From *Microtus montanus fusus* Hall, the subspecies to the south, *M. m. nanus* from Idaho differs as follows: averages smaller; slightly darker and less reddish and less yellowish in color; slightly wider braincase (see measurement of prelamdoidal breadth); larger bullae.

Measurements.—Average (= arithmetical mean) measurements of 34 specimens, both male and female, from several localities in eastern Idaho are: total length, 151; length of tail, 39; hind foot, 19.2; condylobasilar length of the skull, 25.0; zygomatic breadth, 15.0; alveolar length of upper molar tooth-row, 6.4; prelamdoidal breadth, 8.9; and lambdoidal breadth, 11.7.

Average and extreme measurements of six adult males from near Pocatello, Bannock County, Idaho, and nine adult males from near Afton, Lincoln County, Wyoming, are, respectively, as follows: total length, 143(135-150), 163(143-179); length of tail, 35.1(33-38), 42.8(36-49); caudal index, 32.0(28.0-33.1), 35.7(30.6-41.9); hind foot, 18.9(18-20), 18.8(17-20); condylobasilar length of skull, 24.4(24.0-26.0), 25.6(24.5-26.2); alveolobasilar length, 14.1(13.7-14.5), 14.6(13.8-15.0); palatilar length, 13.2(12.9-13.6), 13.8(13.2-14.5); alveolar length of upper molar tooth-row, 6.3(6.1-6.5), 6.3(6.0-6.6); depth of braincase, 7.7(7.5-7.9), 8.0(7.7-8.3); lambdoidal breadth, 11.4(11.0-11.7), 12.0(11.3-12.7); prelamdoidal breadth, 9.1(8.6-9.4), 8.7(8.0-9.4); zygomatic breadth, 14.3(13.8-14.7), 15.3(14.4-16.3); interorbital breadth, 3.6(3.5-3.7), 3.5(3.3-3.7). The average length of the nasal bones in the series from Pocatello is 7.1 mm. The averages, which have not been included in [Table 1](#), for three measurements of the series from Carbon County, Wyoming, are as follows (Encampment, males; Encampment, females; Savery, males; and Savery, females, respectively): alveolobasilar length, 14.4, 14.3, 14.5, 14.3; interorbital breadth, 3.5, 3.4, 3.5, 3.4; depth of braincase, 7.8, 7.6, 7.9, 7.6. Additional measurements are included in [Table 1](#) for other series.

Discussion.—The name *Microtus montanus caryi* Bailey is here placed in synonymy under *M. m. nanus* (Merriam). Vernon Bailey (1917) in his description of *caryi* made four assumptions that have been found to be entirely or partly invalid. First, he assumed that this is an "extreme variant which gradually changes in characters across Nevada and Utah, and reaches its maximum variation in Wyoming." The differences pointed out in subsequent descriptions of subspecies found in the above area do not show a gradual change in any character, or in the number of characters, nor is *caryi* an extreme when compared with the other subspecies. Second, *Microtus nanus* was not, as Bailey assumed, a different species than *Microtus montanus*. Third, he assumed that the characteristics of adults of *nanus* were adequately ascertainable from the thirteen topotypes available to him. Subsequent sampling from Idaho shows that the series of specimens available to Bailey was made up mostly of young and subadult animals. Finally, *caryi* does not occupy as Bailey stated "the meadows along streams in the arid sagebrush country of the Bear River, Green River, and Wind River valleys" exclusively, or characteristically. When the localities from which the species actually is known are plotted, it seems that the arid basin serves as a barrier and that the species is more commonly and abundantly found in montane meadows in the Transition and Canadian life-zones.

Certain samples, here assigned to *M. m. nanus*, that vary from the average of the subspecies deserve comment. For example, mice from the area in Wyoming southwest of the Green River (in the Uinta Mountains) have relatively smaller feet, but are larger in both total length and size of skull. Specimens from near Afton, Lincoln County, Wyoming, are relatively large in both total length and size of skull. This series and specimens from Teton County, Wyoming, are intermediate between *nanus* from Idaho and the newly named subspecies from near Cody, Park County, Wyoming, described below, in terms of both darkness and the amount of reddish color. Mice from Laramie County are more richly reddish-brown. The specimens from near Savery, in Carbon County, Wyoming, are darker. The alveolobasilar length relative to the condylobasilar length is smaller in the series from along Deer Creek, 16 mi. S, 11 mi. W Waltman, Natrona County, Wyoming. The series from the southern tier of counties in Wyoming and some of the specimens from Colorado have relatively wider zygomatic arches. The specimens from southern Sweetwater County,

Wyoming, are relatively paler, have a relatively longer tail and longer hindfeet, lesser condylobasilar length, and wider braincase. Most of these variations are of questionable significance; they may be chance variations owing to errors in sampling.

Much of the south-central part of the state is relatively low and relatively arid. This area includes the arid basin of the Green River and its major tributaries and the arid Red Desert along the continental divide in Sweetwater County. This area might have acted as a barrier to the mice; gene flow might have been prevented between the populations of the western part of the state and those farther east in the Medicine Bow Mountains and Laramie Mountains. Nevertheless geographic variations of subspecific worth have not taken place. The barrier has either not been of as long duration, or has not been so complete and effective, as the other barriers in the state, namely the Absaroka Range, the Big Horn Basin, the Shoshone Basin, and the valley of the North Platte River. These four barriers presumably have led to the differentiation of the two subspecies that are newly named beyond. Each of the two areas which is set apart by these barriers and in which one of the newly named subspecies has evolved is small; therefore there is a lesser amount of suitable habitat available for each of the newly named mice than there is for *M. m. nanus*. It is conceivable, therefore, that in periods of adverse conditions in each of the small areas the size of the effective breeding population may have been so small that random genetic drift could have operated effectively, or that selection was more critical than in a larger, more stable population. It is difficult to test these possibilities because the selective value of the taxonomic characters is unknown. The observed pattern of variation and facts of distribution are, however, not contradictory to the above possibilities.

Specimens examined.—Total, 993, distributed as follows: All specimens unless otherwise indicated are in the University of Kansas Museum of Natural History. Specimens in other museums are labeled as follows: Chicago Natural History Museum (Chi); University of Michigan, Museum of Zoology (Mich); American Museum of Natural History (AMNH); United States National Museum (USNM); Biological Surveys Collection (USBS). Localities that are not represented in [Fig. 1](#) because overlapping or crowding of the symbols

would result are *Italicized*. Localities are arranged from north to south by states, within a state from northwest to southeast by counties, and within a county from north to south.

WYOMING: *Yellowstone Park*: Canyon Camp, 1 (USBS); Lower Geyser Basin, 1 (USBS); Upper Yellowstone River, 2 (AMNH); *North end of Lake, Yellowstone National Park*, 2 (AMNH). *Teton Co.*: Pacific Creek, 1 (USBS); *Big Game Ridge*, 3 (USBS 1, Mich 2); *Whetstone Creek*, 7 (Mich); Moran and environs (6 localities within a 5 mile radius), 28 (USBS 2, Mich 5); *S fork Buffalo River*, 7 (AMNH); 2 mi. W pass, Black Rock Creek, 1 (USBS); *Jenny Lake*, 5 (Mich); Bar BC Ranch, 2½ mi. NE Moose, 6500 ft., 2; Teton Pass above Fish Creek, 1 (USBS); Jackson and environs, 142 (Mich 141); *Sheep Creek*, 2 (Mich). *Lincoln Co.*: 13 mi. N, 2 mi. W Afton, 2; 10 mi. N, 2 mi. W Afton, 4; 9½ mi. N, 2 mi. W Afton, 3; 9 mi. N, 2 mi. W Afton, 9; 7 mi. N, 1 mi. W Afton, 12; Afton, 1 (USBS); Labarge Creek, 1 (USBS); Border, 6 (USBS); *Cokeville*, 2 (USBS); 6 mi. N, 2 mi. E Sage, 1; Cumberland, 5 (USBS). *Sublette Co.*: 34 mi. N, 4 mi. W Pinedale, 1; 33 mi. N, 2 mi. W Pinedale, 6; 32 mi. N, 1 mi. W Pinedale, 1; 31 mi. N Pinedale, 4; Dell Creek, on Ferris Ranch, 7 (Mich); Horse Creek, 7800 ft., Merna, 4 (USBS); Big Piney, 1 (USBS). *Fremont Co.*: 17½ mi. W, 2½ mi. N Lander, 9500 ft., 3; 17 mi. W, 2 mi. N Lander, 9300 ft., 4; Milford and environs (5 localities within a 1 mile radius), 23 (USBS 4); 15½ mi. S, 7½ mi. W Lander, 9200 ft., 1; South Pass City, 8000 ft., 8 (USBS); 23½ mi. S, 5 mi. W Lander, 8600 ft., 7. *Natrona Co.*: Deer Creek, 16 mi. S, 11 mi. W Waltman, 6950 ft., 44; 6 mi. S, 2 mi. W Casper, 5900 ft., 4; 6-4/5 mi. S, 2 mi. W Casper, 6100 ft., 1; 7 mi. S, 2 mi. W Casper, 6370 ft., 3; 10 mi. S Casper, 7750 ft., 33; Sun, 2 (USBS); 5 mi. W Independence Rock, 6000 ft., 4; 5 mi. W, 1 mi. S Independence Rock, 2. *Converse Co.*: Beaver, 1 (USBS). *Uinta Co.*: 1½ mi. W, ½ mi. S Cumberland, 6; 16 mi. S, 2 mi. W Kemmerer, 6700 ft., 3; 10 mi. SW Granger, 3 (Mich); Fort Bridger, 6650 ft., 25 (USNM 6); 9 mi. S Robertson, 8000 ft., 9; 9½ mi. S, ½ mi. W Robertson, 8600 ft., 1; 10 mi. S, 1 mi. W Robertson, 8700 ft., 25; 14 mi. S, 2 mi. E Robertson, 9000 ft., 5; 4 mi. S Lonetree, 1 (USBS). *Sweetwater Co.*: Farson, 3; Bitter Creek, 3 (AMNH); Kinney Ranch, 21 mi. S Bitter Creek, 6800 ft., 9 (USNM 1, AMNH 2); 32 mi. S, 22 mi. E Rock

Springs, 7025 ft., on Vermillion Creek, 15. *Carbon Co.*: 18 mi. NNE Sinclair, 6500 ft., 10; Bridgers Pass, 18 mi. SW Rawlins, 7500 ft., 7; Saratoga, 1 (USBS); 6 mi. S, 13 mi. E Saratoga, 8500 ft., 5; 6 mi. S, 14 mi. E Saratoga, 8800 ft., 1; Lake Marie, 10,440 ft., 2; 1 mi. S Lake Marie, 2; ½ mi. S, 2 mi. E Medicine Bow Peak, 10,800 ft., 1; Encampment (12 localities from 10 mi. N, 14 mi. E to 9 mi. N, 3 mi. E Encampment and from 6500 to 8400 ft.), 63; ¼ mi. N Riverside, 7380 ft., 2; S base Bridger Peak, 8800 ft., Sierra Madre Mountains, 1; 2 mi. S Bridger Peak, 9300 ft., 2; Savery (10 localities from 8 mi. N, 21 mi. E to 4 mi. N, 8 mi. E Savery and from 7300 to 8800 ft.), 80. *Albany Co.*: 30 mi. N, 10 mi. E Laramie, 6760 ft., 6; 29¾ mi. N, 9½ mi. E Laramie, 6350 ft., 1; 26 mi. N, 4½ mi. E Laramie, 6960 ft., 8; 26¾ mi. N, 6½ mi. E Laramie, 6700 ft., 3; 3 mi. N, 13 mi. E Laramie, 7500 ft., 1; 7 mi. N, 2 mi. E Laramie, 1 (Chi); 5 mi. N Laramie, 7400 ft., 15; Laramie, 4 (AMNH); 1 mi. E Laramie, 7160 ft., 4; 7-7/10 mi. SSW Laramie, 7200 ft., 4; 6½ mi. S, 8¾ mi. E Laramie, 8200 ft., 1; Headquarters Park, 10,200 ft., Medicine Bow Mountains, 3 (USBS); Centennial, 8120 ft., 1; 2-¼ mi. ESE Brown's Peak, 10,300 ft., 3; 3 mi. ESE Brown's Peak, 10,000 ft., 12; 2 mi. S Brown's Peak, 10,600 ft., 1; Pole Mountain, 15 mi. SE Laramie, 4 (USBS 3); 1 mi. SSE Pole Mountain, 8350 ft., 4; 2 mi. SW Pole Mountain, 8300 ft., 13; 3 mi. S Pole Mountain, 8100 ft., 1; Sherman, 2 (AMNH). *Laramie Co.*: 5 mi. N, 1 mi. W Horse Creek P. O., 7200 ft., 1; Meadow, 2 (USBS); 11 mi. N, 5½ mi. E Cheyenne, 5450 ft., 7; 7 mi. W Cheyenne, 6500 ft., 10; Cheyenne, 3 (USNM).

COLORADO: *Moffat Co.*: Lay, 6160 ft., 1 (AMNH). *Routt Co.*: Wright's Ranch, Yampa, 7700 ft., 2; Gore Range, 8 mi. E Toponas, 8000 ft., 2 (USBS). *Larimer Co.*: 12½ mi. W, 1½ mi. S Rustic, 1; 11 mi. W, 1 mi. S Rustic, 1; Cache La Poudre River, 1 (Chi); *Estes Park*, 3 (USBS 1, AMNH 2); 19½ mi. W, 2½ mi. S Loveland, 7280 ft., 6; 16 mi. W Loveland, 6840 ft., 1; 6 mi. W, ½ mi. S Loveland, 5200 ft., 1. *Rio Blanco Co.*: Meeker, 1 (USBS); 9½ mi. SW Pagoda Peak, 7700 ft., 3; 5 mi. S Pagoda Peak, 9100 ft., 2. *Eagle Co.*: Eagle, 1 (USBS); Pando, 2 (USBS). *Grand Co.*: Mt. Whiteley, 2 (USBS); Arapahoe Pass, Rabbit Ear Mountains, 2 (USBS); Coulter (near Granby), 5 (USBS); Arrowhead (near Dale), 1 (USBS). *Boulder Co.*: ¾ mi. N, 2 mi. W

Allenspark, 8400 ft., 4; 3 mi. S Ward, 9000 ft., 3; Nederland, 16 (Chi). Clear Creek Co.: Mt. McLellan, 2 (USBS); Berthoud Pass, 4. Park Co.: Trout Creek Ranch, 2 mi. N Garo, 1 (USBS).

Specimens examined of *M. m. nanus* from eastern Idaho and Montana are as follows: IDAHO: Custer Co.: Challis, 7 (USBS); Mill Creek, Challis Nat. Forest, 1 (USBS); Pahsimeroi Mts., 12 (USBS); Lost River Mts., 1 (USBS). Fremont Co.: N fork Snake River, 10 mi. SW Island Park, 6200 ft., 2 (AMNH); Black Springs Creek, 4 mi. W Ashton, 5200 ft., 1 (AMNH); 5 mi. W St. Anthony, 5000 ft., 1 (AMNH). Camas Co.: Camas Prairie, Corral, 5100 ft., 2 (USBS). Blaine Co.: Alturas Lake, 3 (USBS); Sawtooth Lake, 2 (USBS); Craters of the Moon, Laidlow Park, 2 (Mich); Ticura, 10 mi. S Picabo, 1 (USBS); 19 mi. NE Carey (Lava Lake), 8 (Mich). Butte Co.: 26 mi. SW Arco, 12 (Mich). Bingham Co.: Shelley, 6 (USBS). Bonneville Co.: 10 mi. SE Irwin, 4 (USBS). Owyhee Co.: Three Creeks, 3 (USBS). Twin Falls Co.: Castleford Fenced Plot, 11 mi. W, 9 mi. S Twin Falls, 1. Minidoka Co.: Heyburn, 2 (USBS). Cassia Co.: 2 mi. S, 2 mi. W Burley, 5. Bannock Co.: Pocatello, 23 (USBS 4); Swan Lake, 1 (USBS). Bear Lake Co.: Montpelier Creek, 6700 ft., 3 (USBS). MONTANA: Gallatin Co.: W. Fork of W. Fork, Gallatin River, 1 (USBS). Park Co.: Lamar River, 7000 ft., 1 (USBS); Gardiner, 1 (USBS). Sweet Grass Co.: 14 mi. S Big Timber, 1 (USBS); McLeod, 1 (USBS); West Boulder Creek, 18 mi. SE Livingston, 2 (USBS).

***Microtus montanus codiensis*, new subspecies**

Type.—Female, adult, skin and skull; No. 27578, Museum of Natural History, University of Kansas, from 3½ mi. E and ¾ mi. S Cody, 5020 ft., Park Co., Wyoming; obtained on August 11, 1948, by James W. Bee, original number 18-8-11-48.

Range.—In northwestern Wyoming eastward from the Absaroka and Wind River ranges into the western part of the Big Horn Basin.

Diagnosis.—A relatively large *Microtus montanus*; tail actually and relatively long; hind foot actually but not relatively large; skull large; zygomatic expanse actually and relatively large; alveolobasilar length

relatively large; upper molar tooth-row relatively long; color relatively light, not reddish.

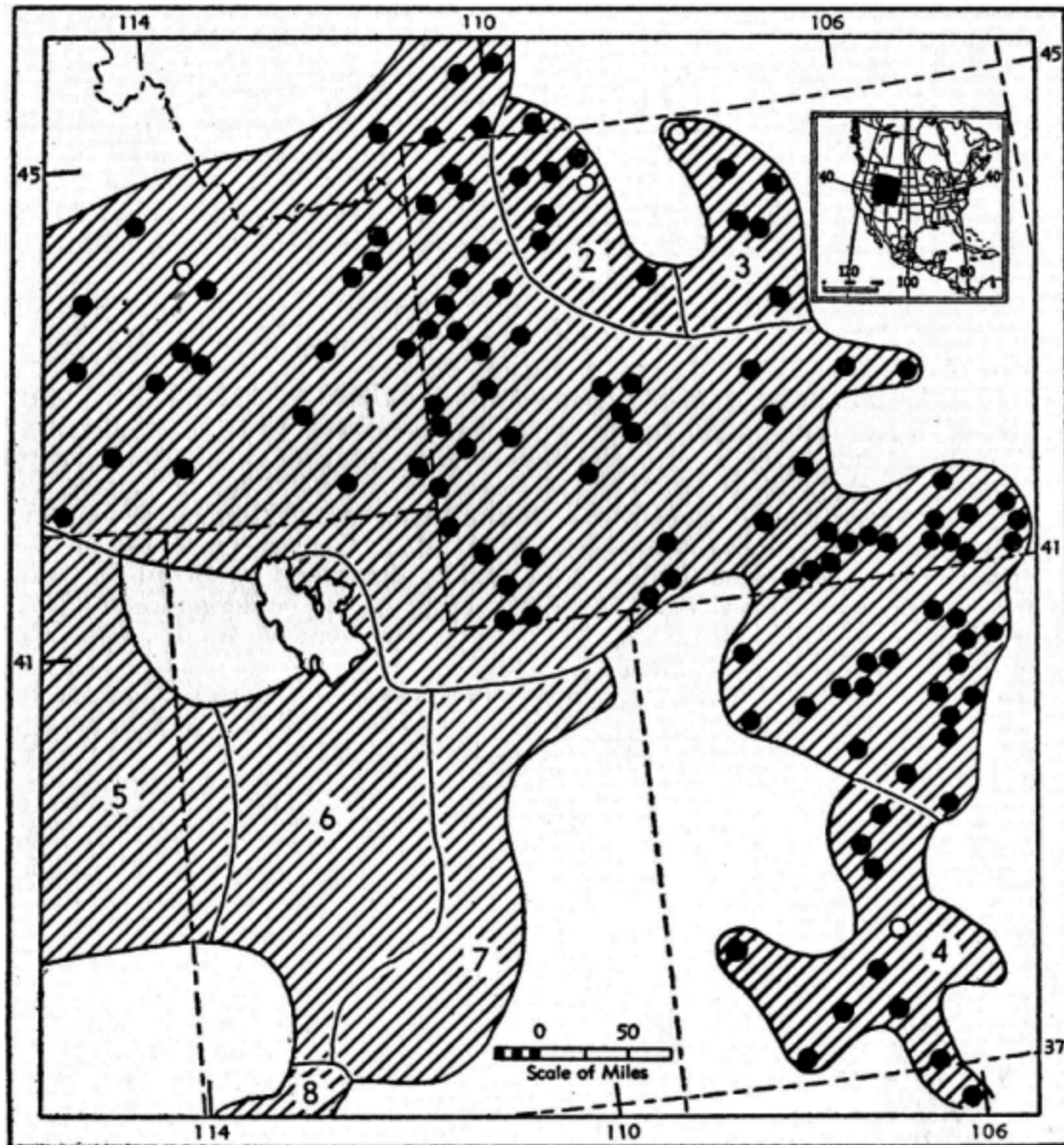


Fig. 1. Geographic range of *Microtus montanus* in Wyoming, Colorado, and adjacent areas. The solid circles represent localities from which specimens have been examined; the hollow circles represent type localities. The ranges of subspecies in Utah are after Durrant, 1952.

Guide to subspecies

1. *M. m. nanus*
 2. *M. m. codiensis*
 3. *M. m. zygomaticus*
 4. *M. m. fusus*
 5. *M. m. micropus*
 6. *M. m. nexus*
 7. *M. m. amosus*
 8. *M. m. rivularis*
-

Comparisons.—As compared with the specimens of *M. m. nanus* from Idaho, the size is larger (see diagnosis and measurements). Certain proportions which differ from those of *nanus* and which are not in close agreement with the observed differences with age in specimens of *nanus* of a size comparable to *codiensis* are relatively large alveolobasilar length, relatively long alveolar length of upper molar tooth-row, relatively wide-spreading zygomatic arches, and relatively long tail. The color in *codiensis* is lighter than in *nanus*. As compared to the new subspecies named below from the Big Horn Mountains to the east, *codiensis* is of similar size in head-body length, but has a relatively as well as actually longer tail; the hind foot averages longer; the upper molar tooth-row is relatively longer; the color is slightly paler and less grizzled; the bullae are larger and less flattened; the angle formed at the suture between the basioccipital and basisphenoid bones is less acute; and the region of the suture is less prominently elevated between the bullae when viewed from the ventral aspect. The pterygoid plates mesial and posterodorsal to the posterior end of the last upper molar are less fenestrated, and the incisive foramina are less constricted posteriorly.

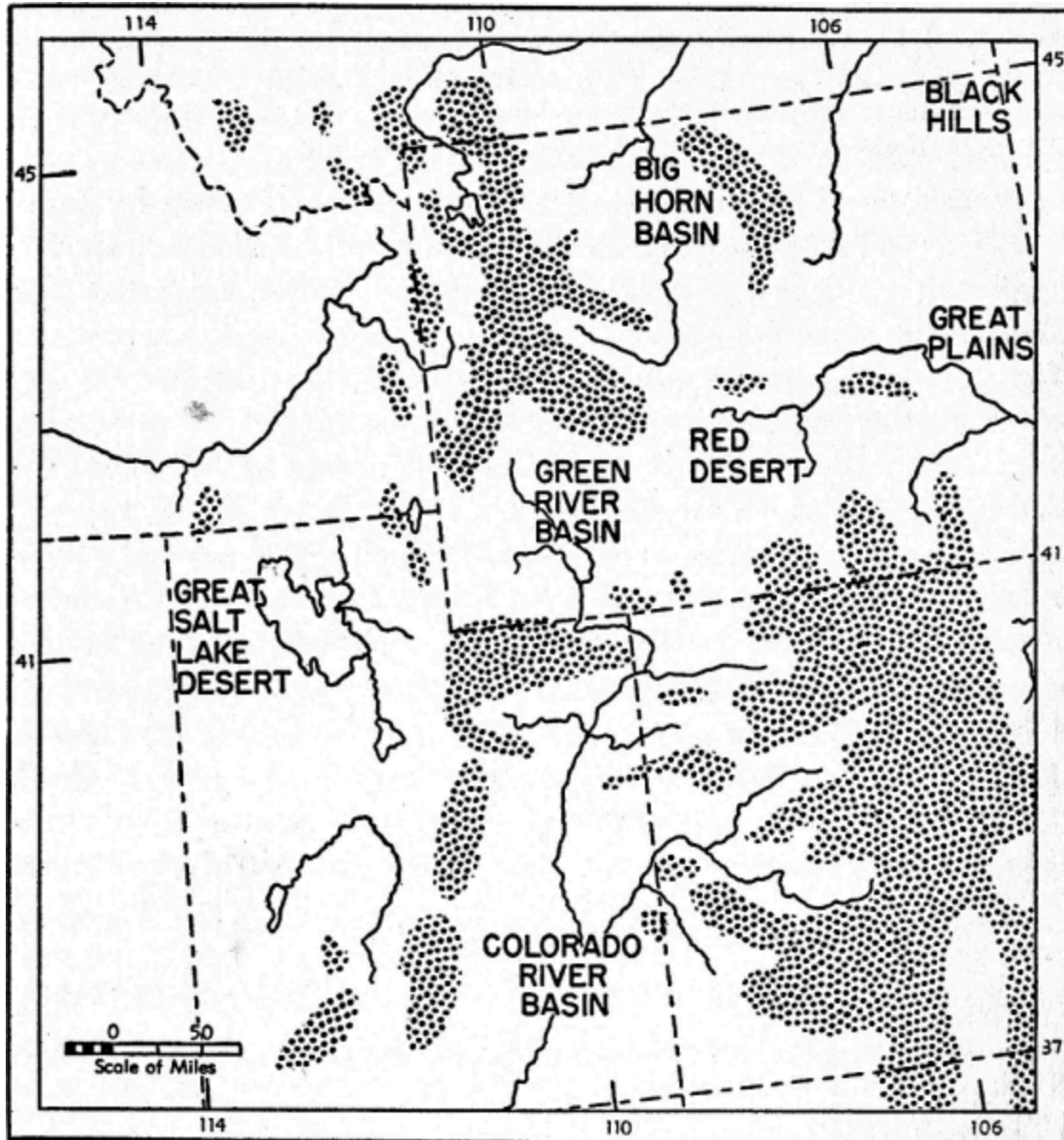


Fig. 2. Map showing the major barriers to *Microtus montanus* in Wyoming and Colorado; the barriers are the low areas named on the map (the name "Black Hills" is on the map for another reason; these hills are not a barrier). The major mountainous areas higher than approximately 8000 feet in elevation in Wyoming, Colorado and Utah are stippled. These mountainous areas include the habitat that is most suitable for the montane meadow mouse. The Black Hills are unoccupied by this species but these hills seem to be ecologically suitable for the species.

Measurements.—The average and the extremes for some measurements of 34 males and females, 27 from the type locality and 7 from other localities in the range assigned to this subspecies, are as follows: total length, 165 (146-186); length of tail, 44.2 (35-55); hind foot, 19.6 (17-21); condylobasilar length of the skull, 25.5 (24.0-27.5); zygomatic breadth, 15.6 (14.7-16.6); alveolar length of upper molar tooth-row, 6.6 (6.2-7.0); prelamdoidal breadth, 8.8 (8.1-9.5); lambdoidal breadth, 12.0 (11.2-12.8). As an indication of variability and for comparison with other series the coefficient of variability and two times the standard error of the mean for each measurement in this series are included in [Table 1](#). The averages for some measurements of 27 topotypes are as follows: total length, 162; length of tail, 45.5; hind foot, 19.9; condylobasilar length, 25.6; palatilar length, 14.0; molar series, 6.6; alveolobasilar length, 14.9; zygomatic breadth, 15.6; interorbital breadth, 3.5; lambdoidal breadth, 12.1; prelamdoidal breadth, 8.9; depth of braincase, 7.8.

Discussion.—Three species of *Microtus* were collected by James W. Bee at the type locality. *Microtus montanus codiensis*, *Microtus longicaudus mordax*, and *Microtus pennsylvanicus modestus* were taken in the same runways in the same meadow, at the same time. *Microtus ochrogaster haydeni*, although not taken at this locality, occurs in the Big Horn Basin. These four species differ in their geographic ranges, being largely allopatric, except *M. montanus* and *M. longicaudus* which are sympatric. Although the different species have ecological preferences and habits which differ, several species of *Microtus* may occur together in local areas such as the above. Certain of the characteristics of *M. m. codiensis* are intermediate between those of the species *M. montanus* on one hand and those of the other three species on the other hand. Could interspecific hybridization between "good species" of *Microtus* take place in nature and possibly alter the characteristics of a local population?

Specimens examined.—Total, 50, distributed as follows (abbreviations for collections are given in the account of *M. m. nanus*; localities that are not represented in [Fig. 1](#) because overlapping or crowding of the symbols would result are italicized):

MONTANA: *Carbon Co.*: Beartooth Mountains, 2 (USBS); *Beartooth Lake*, 1 (USBS).

WYOMING: *Park Co.*: Black Mountain, head of Pat O'Hara Creek, 3 (USBS); 13 mi. N, 1 mi. E Cody, 5200 ft., 1; SW slope Whirlwind Peak, 9000 ft., 1; 5 mi. N Cody, 6300 ft., 1 (USBS); 3½ mi. E, ¾ mi. S Cody, 31; Ishawooa Creek, 6300 ft., 2 (USBS); *Valley*, 1 (USBS); Needle Mountain, 10,500 ft., 4 (USBS). *Hot Springs Co.*: 3 mi. N, 10 mi. W Thermopolis, 4950 ft., 3.

***Microtus montanus zygomatikus*, new subspecies**

Type.—Male, adult, skin and skull, No. 32761, Museum of Natural History, University of Kansas, from Medicine Wheel Ranch, 9000 ft., 28 mi. E Lovell, Big Horn County, Wyoming; obtained by R. Freiburg, original number 105.

Range.—The Big Horn Mountains of north-central Wyoming.

Diagnosis.—A large *Microtus montanus* with a relatively short tail; short molar series; broad zygomatic arches well rounded in lateral outline when viewed from above; small and flattened bullae; raised basioccipito-basisphenoid suture.

Comparisons.—For comparison with *M. m. codiensis* from the west, on the other side of the Big Horn Basin, see the account of that subspecies. In comparison with *nanus* this subspecies is slightly paler, in this respect showing more resemblance to *codiensis* although not so pale, and more grizzled or unevenly colored. This difference in color between *zygomatikus* and *codiensis* may not be of taxonomic significance. From both the topotypes of *nanus*, and the series of it from Wyoming, *zygomatikus* differs on the average in having a relatively shorter tail, a relatively shorter upper molar tooth-row, relatively more rounded and relatively more wide-spread zygomatic arches, and smaller more flattened bullae.

Measurements.—Average and extreme measurements of 24 adult males and females from several localities here referred to *M. m. zygomatikus* are as follows: total length, 159(150-175); length of tail,

37.6(31-46); hind foot, 18.6(17-20); condylobasilar length of the skull, 25.3(24.2-26.7); zygomatic breadth, 15.3(14.1-16.7); alveolar length of upper molar tooth-row, 6.2 (5.7-6.8); prelamdoidal breadth, 8.7(8.3-9.4); lambdoidal breadth, 11.9(11.0-12.5). Average and extreme measurements of a series of 12 adult male topotypes are as follows: total length, 159(144-174); length of tail, 36.4 (30-41); hind foot, 18.2(16-20); condylobasilar length of skull, 25.8(24.7-26.7); alveolobasilar length, 14.8(13.8-15.3); palatilar length, 13.8 (12.7-14.2); alveolar length of upper molar tooth-row, 6.4(5.9-6.6); zygomatic breadth, 15.9 (15.0-16.7); interorbital breadth, 3.6(3.4-3.7); lambdoidal breadth, 12.1 (11.5-12.5); prelamdoidal breadth, 8.6(8.3-8.9); depth of braincase, 8.0 (7.6-8.3).

Discussion.—This subspecies is separated from *M. m. codiensis* to the west by the Big Horn Basin. A series from along Buffalo Creek, 27 mi. N, 1 mi. E Powder River, 6075 ft., in Natrona County, Wyoming, is intermediate between the topotypes of *zygomaticus* and *nanus* in the characters cited above as distinguishing the two, but shows greater resemblance to *zygomaticus* in the shape of the zygomatic arch, in color which is paler than in topotypes of *zygomaticus*, and in the short hind foot. On these and on geographic grounds this population is referred to *zygomaticus*. Unfortunately we cannot be certain in many cases that an intermediate condition in a certain character indicates a genetically intermediate population and therefore true intergradation between the two subspecies to which the population is geographically intermediate. The topotypes of this subspecies are the most distinct of all the series which I have studied from the eastern Rocky Mountains, in terms of the degree of morphological departure from the norm for the species. After *zygomaticus* the following populations are arranged according to their degree of deviation from this norm (*codiensis* deviates most): topotypes of *codiensis*, *fusus* and a population from southern Sweetwater County, Wyoming, and lastly the *nanus-caryi* complex. Within the latter group, as I have mentioned, there are a number of local variants most of which do not differ significantly and do not conform to any geographic pattern.

Specimens examined.—Total, 55, distributed as follows (abbreviations for collections are given in the account of *M. m. nanus*; localities that

are not represented in [Fig. 1](#) because overlapping or crowding of the symbols would result are Italicized): WYOMING: *Big Horn Co.*: Medicine Wheel Ranch, 9000 ft., 28 mi. E Lovell, 30; W slope, head of Trappers Creek, 9500 ft., 2 (USBS). *Washakie Co.*: 9 mi. E, 5 mi. N Tensleep, 7400 ft., 1. *Johnson Co.*: 7½ mi. W, 1 mi. S Buffalo, 6500 ft., 3; Big Horn Mountains, 3 (USBS). *Natrona Co.*: Buffalo Creek, 27 mi. N, 1 mi. E Powder River, 6075 ft., 16.

***Microtus montanus fusus* Hall**

Microtus nanus, Bailey, N. Amer. Fauna, 17:30, June 6, 1900 (part); Cary, N. Amer. Fauna 33:123, August 17, 1911.

Microtus montanus fusus Hall, Proc. Biol. Soc. Washington, 51:131-134, August 23, 1938; Warren, The Mammals of Colorado, Univ. of Okla. Press, p. 229, 1942.

Type.—Male, adult, skin and skull; No. 61281, Museum of Vertebrate Zoology; 2½ miles east of summit of Cochetopa Pass, Saguache County, Colorado; Sept. 21, 1933; collected by Annie M. Alexander; original number 2568. Type not seen by me.

Range.—Southern Colorado and northern New Mexico.

Comparisons.—For comparison with *M. m. nanus*, the subspecies to the northward, see the preceding account of that subspecies. For comparison with *M. m. amosus* the subspecies to the west see Hall (1938) and Durrant (1952). I have not examined specimens of *amosus*.

Measurements.—Average and extreme measurements for 17 adults including both males and females from several localities in southern Colorado are as follows: total length, 160 (136-179); length of tail, 42 (35-55); hind foot, 19.2 (17-23); condylobasilar length of the skull, 25.2 (24.0-26.0); zygomatic breadth, 15.0 (14.1-15.5); alveolar length of upper molar tooth-row, 6.4 (6.0-6.7); prelambdoidal breadth, 8.7 (8.3-9.2); lambdoidal breadth, 11.7 (11.1-12.6).

Average and extreme measurements of 4 adults (2 males and 2 females) from the type locality and 11 adults (4 males and 7 females)

from other localities in southern Colorado are as follows: total length, 162 (157-168), 157 (137-169); length of tail (means only), 44.5, 40.5; hind foot, 18.8 (18-19), 18.6 (18-23); condylobasilar length of skull, 24.5 (24.0-24.7), 25.2 (24.3-26.1); alveolobasilar length, 14.2 (13.9-14.5), 14.6 (14.1-15.1); palatilar length, 13.2 (13.0-13.4), 13.5 (13.1-14.2); alveolar length of upper molar tooth-row, 6.3 (6.0-6.6), 6.4 (6.3-6.7); zygomatic breadth, 15.0 (14.3-15.5), 14.9 (14.1-15.5); interorbital breadth, 3.5 (3.3-3.6), 3.5 (3.3-3.7); lambdoidal breadth, 11.8 (11.1-12.6), 11.7 (11.2-12.3); prelambdoidal breadth, 8.6 (8.3-9.2), 8.8 (8.3-9.0); depth of braincase, 7.5 (7.2-7.8), 7.6 (7.1-7.9).

Discussion.—There is no sharp boundary between *M. m. fusus* of southern Colorado and the subspecies to the north, *M. m. nanus*. Although the line separating these two subspecies is drawn somewhat arbitrarily, on the whole the samples from north of this line more closely resemble *nanus*. All of the means for total length given above are larger than the maximum given in Hall's description of *fusus*. The caudal index (38 and 35% in two series) is slightly larger than that cited by Hall (33.3%) and is not significantly different from that in *nanus* (35.2%). The color in both young and old mice is variable, but in general is more yellowish, and less grayish, than in any other series studied.

There is a large area in western Colorado and eastern Utah, between the known ranges of *M. m. fusus* and *M. m. amosus* from which there are no specimens. Probably the species occurs only at certain places in this arid region which seems to be a partial barrier to the species.

Specimens of *M. montanus* from northern New Mexico have been referred previously to *M. m. arizonensis*. When he named *M. m. fusus*, Hall mentioned its resemblance to *arizonensis* in reddish coloration, but pointed out that *fusus* is less reddish. Of six specimens from Valle Santa Rosa, Jemez Mountains (USBS), 8500 ft., Rio Arriba County, New Mexico, three are immature, and the skulls of the remaining specimens are damaged. In reddish color and relatively large size these few specimens resemble *arizonensis* more than *fusus* although the locality of occurrence is closer to the geographic range of the northern *fusus* than to that of *arizonensis*. The identification of these specimens as *arizonensis* is provisional; additional specimens are needed from the area, 200 miles wide, which separates the

ranges as now known of *arizonensis* in Arizona from the occurrence in New Mexico. There is a single specimen from this area, the damaged skull of which prevents conclusive identification. The specimen is either *M. montanus* or *M. mexicanus*, and is from Nutria, on the southern edge of the Zuni Mountains (USBS). Detailed comparison of *fusus* and *arizonensis* is not attempted here although it may be stated that in several characters *fusus* is intermediate between *arizonensis* to the south and *nanus* to the north.

Specimens examined.—Total, 89, distributed as follows (abbreviations for collections are given in the account of *M. m. nanus*; localities that are not represented in [Fig. 1](#) because overlapping or crowding of the symbols would result are Italicized):

COLORADO: *Pitkin Co.*: 5 mi. W Independence Pass, 11,000 ft., 1 (Chi). *Lake Co.*: *Independence Pass*, 12,095 ft., 2 (Chi). *Gunnison Co.*: *Gothic*, 2 (USBS); *Decker's Ranch*, *Crested Butte*, 2 (AMNH); *Almont*, 3 (USBS). *Montrose Co.*: *Coventry*, 5 (USBS 4, AMNH 1). *Saguache Co.*: *Cochetopa Pass and environs*, 44 (USBS 22). *Hinsdale Co.*: *Ruby Lake*, 1 (USBS). *Mineral Co.*: 3 mi. E Creede, 1; 23 mi. S, 11 mi. E Creede, 9300 ft., 7. *La Plata Co.*: *Florida*, 6800 ft., 1. *Conejos Co.*: 1 mi. S, 19 mi. W Antonito, 10,200 ft., 3; 4 mi. S, 23 mi. W Antonito, 1; 5 mi. S, 24 mi. W Antonito, 9600 ft., 9.

NEW MEXICO: *Rio Arriba Co.*: 6 mi. W Hopewell, 9900 ft., 6 (USBS); *Tusas River*, 8700 ft., 1 (USBS).

Some measurements not given above are included in [Table 1](#), together with the number of specimens and the sex if restricted to one sex. So that the variability can be evaluated more adequately, the coefficient of variability and 2 times the standard error of the mean are included for the measurements in two series. The series consist of all the adult specimens (with a condylobasilar length of 24.0 mm. or more) of both sexes from the areas specified. Various barriers are shown in [Fig. 2](#) for comparison with the distributions of the subspecies and the localities of known occurrence shown in [Fig. 1](#). *Microtus montanus* has not been taken in the Black Hills area of extreme northeastern Wyoming. Suitable montane habitat is present and both *Microtus pennsylvanicus insperatus* and *Microtus longicaudus*

longicaudus occur there. The arid basin of the Powder River presumably is a barrier that has prevented *M. montanus* from reaching this area.

TABLE 1. AVERAGE MEASUREMENTS, IN MILLIMETERS, OF ADULTS OF *MICROTUS MONTANUS*.

Key to column headings:

A: No. of individuals averaged
 B: Total length
 C: Length of tail
 D: Length of hind foot
 E: Condylbasilar length
 F: Alveolar length of upper molar tooth-row
 G: Zygomatic breadth
 H: Lambdoidal breadth
 I: Prelambdoidal breadth

LOCALITY	A	B	C	D	E	F	G	H	I
<i>M. m. codiensis</i> , all									
Average	34	165.3	44.2	19.6	25.4	76.5	615.5	512.0	8.76
2 × stand. error		3.5	61.8	4.3	95.3	08.0	67.1	98.1	.144
Coeff. variab		6.0	11.6	5.6	3.5	3.0	3.65	1.20	1.47
<i>M. m. nanus</i> , Eastern Idaho									
Average	21 ^[1]	151.1	39.4	19.2	25.0	6.44	14.99	11.74	8.94
2 × stand. error		3.2	02.8	9.2	93.2	861.1	5.29	5.21	0.182
Coeff. variab.		6.1	21.1	4.3	8.2	493.9	94.1	3.79	4.31
<i>M. m. nanus</i> , Wyoming									
Teton Co.	35	160.5	40.7	18.6	25.1	66.5	15.17	11.86	8.77
Fremont Co.	26	157.0	41.4	19.6	25.2	36.2	515.0	511.8	8.91
Lincoln Co.	24	159.9	41.8	18.9	25.0	86.2	615.1	1011.8	28.75
Uinta Co.	26	162.4	41.3	19.0	25.3	36.4	215.3	1212.1	68.89
Sweetwater Co.	12	159.8	43.7	20.1	24.9	86.3	15.00	11.84	9.02

Natrona Co.	40	159.6	41.0	19.6	25.0	6.40	15.00	11.8	48.93
Carbon Co.	108	158.7	40.0	19.1	24.9	6.27	15.05	11.8	38.72
Encampment ♂	27	161	41.7	18.9	25.1	6.16	15.2	11.9	8.7
Encampment ♀	11	159	41.1	19.4	24.9	6.18	14.9	11.6	8.6
Savery ♂	23	159	41.0	19.2	25.3	6.3	15.2	12.2	8.8
Savery ♀	25	155	37.1	18.8	24.7	6.3	14.8	11.6	8.6
<i>M. m. nanus</i>									
Northern Colo.	8	163.1	42.4	19.6	25.2	6.44	14.86	11.7	08.56
<i>M. m. fusus</i>									
Southern Colo.	17 ^[2]	159.8	42.4	19.2	24.9	6.43	14.98	11.7	38.69

^[1] For external parts, 34 individuals were used.

^[2] For external parts, 29 individuals were used.

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Transcriber's Notes

One typographical error was corrected: "Castlford" was corrected to "Castleford" in "Twin Falls Co.: Castleford Fenced Plot".

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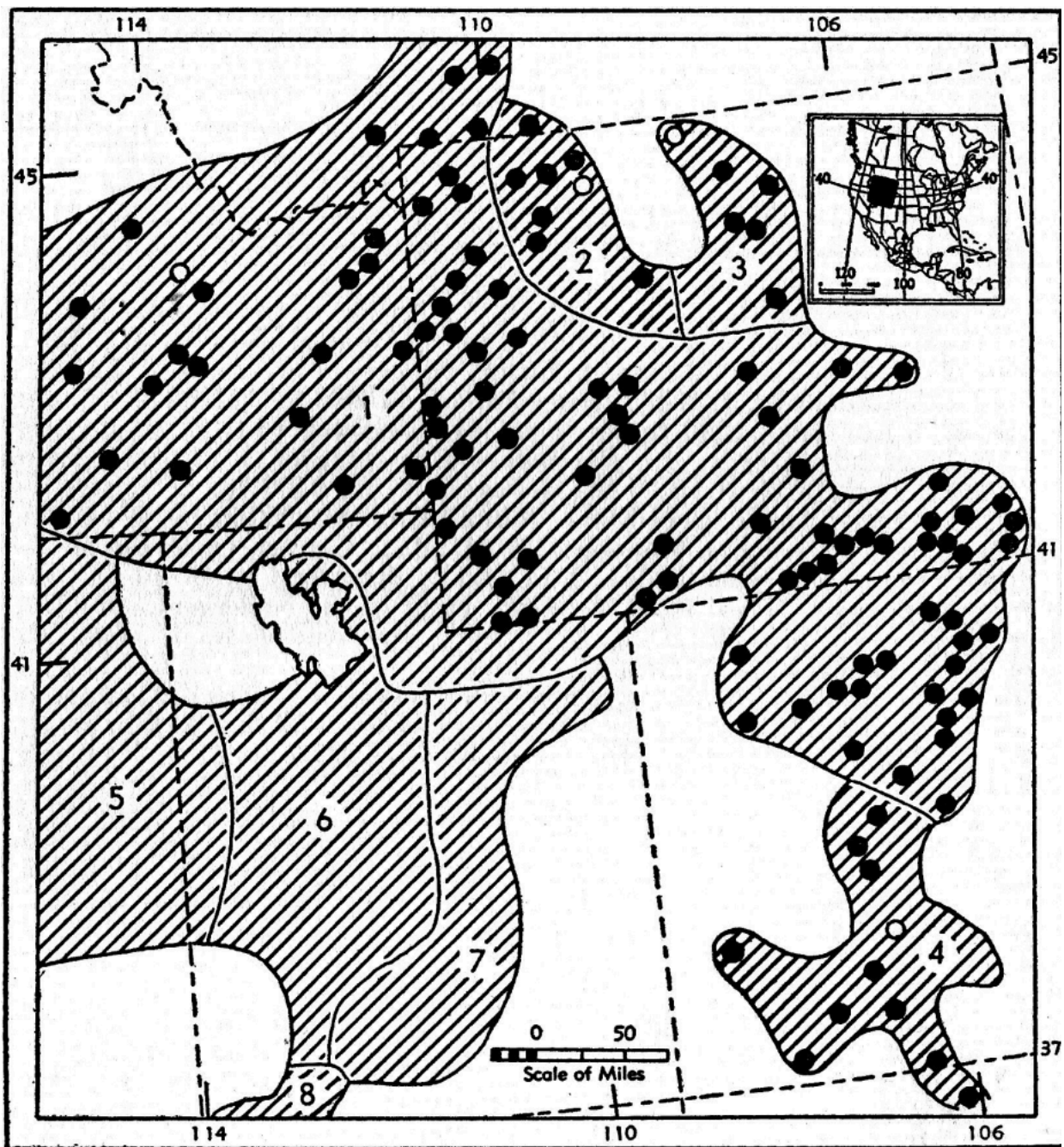
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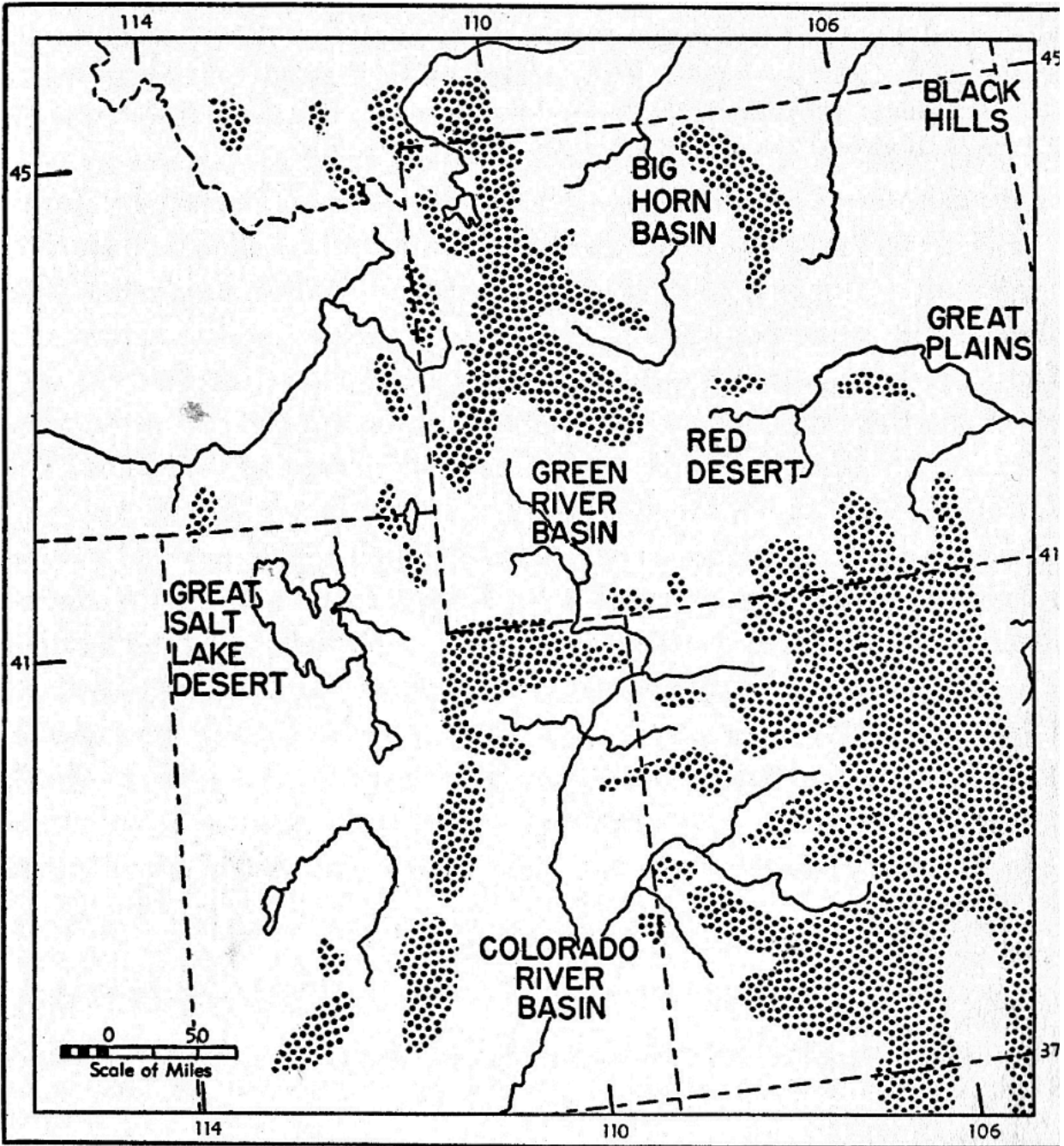
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